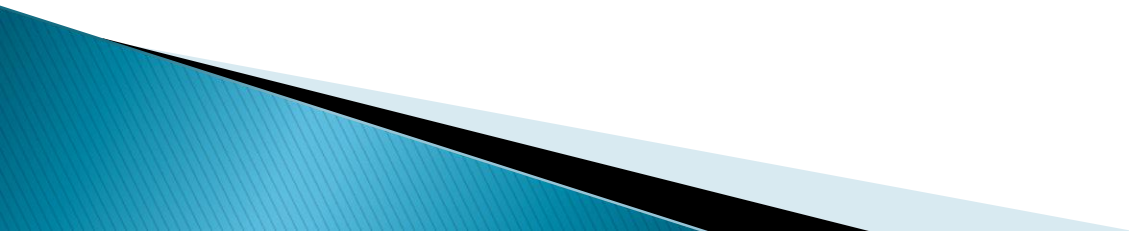


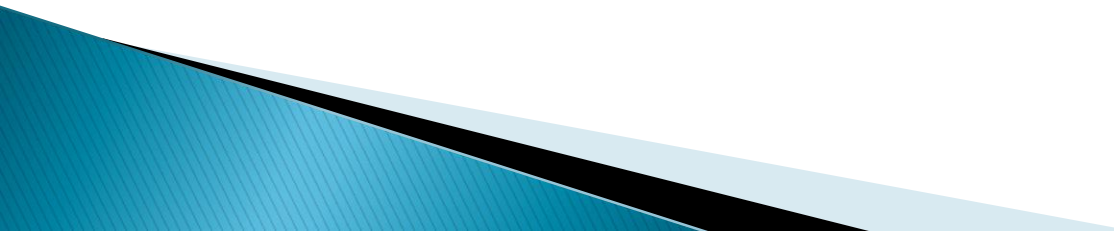
SOFTWARE QUALITY ASSURANCE (SQA)



Software Quality

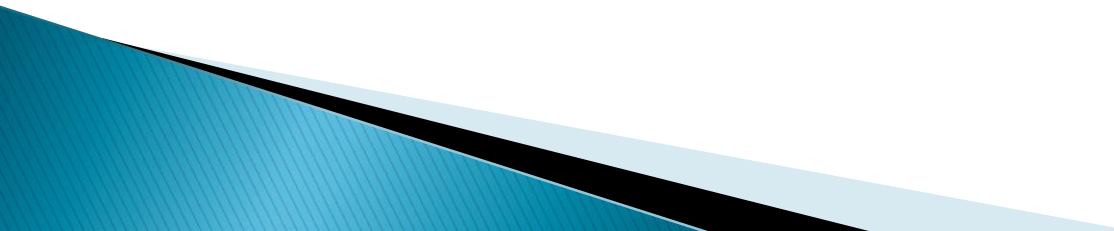
Definition –Software Quality

Conformance to

- ▶ 1) Explicitly stated functional and performance requirements
 - ▶ 2) Explicitly documented development standards
 - ▶ 3) Implicit characteristics that are expected of all professionally developed s/w
- 

Software Quality

Emphasizes

- ▶ Software requirements are the foundation from which quality is measured. Lack of conformance to requirement is lack of quality.
 - ▶ Standards guides the manner in which s/w is engineered. If criteria are not followed then lack of quality ensues.
 - ▶ If s/w conforms to explicit requirements but fails to meet implicit requirements s/w quality is suspect.
- 

Mc Calls Quality Factors

Focuses on 3 important aspects of a s/w product

1. Its operational characteristics (**Product Operation**)
2. Its adaptability to undergo change (**Product revision**)
3. Its adaptability to new environment. (**Product transition**)

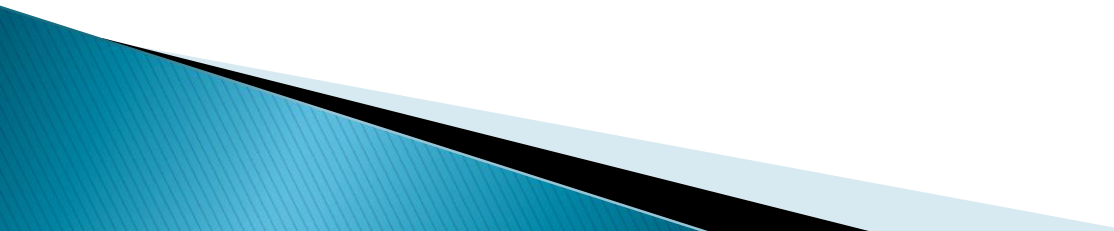
Factors :

1. **Correctness** :Extent to which a program satisfies its specification and fulfills customer objectives
 2. **Reliability** :Extent to which a program can be expected to perform its intended function with required precision.
- 

Mc Calls Quality Factors

- 3) **Efficiency:** The amount of computing resources and code required by a program to perform its function.
 - 4) **Integrity :** Extent to which access to s/w or data by unauthorized persons can be controlled.
 - 5) **Usability:** Effort required to learn operate, prepare input and interpret operation of a program
 - 6) **Maintainability:** Effort required to locate and fix an error in a program.
 - 7) **Flexibility:** Effort required to modify operational program.
 - 8) **Testability:** Effort required to test program to ensure that it is performing its intended function.
 - 9) **Portability:** Effort required to transfer the program from one s/w or h/w system to another.
 - 10) **Reusability:** Extent to which programs can be reused in other applications.
 - 11) **Interoperability:** Effort required to couple one system to another.
- 

Mc Calls Quality Factors

- ▶ **PRODUCT OPERATION** :Factors which are related to operation of a product
 - ▶ **PRODUCT REVISION** :Factors which are required for testing and maintainence
 - ▶ **PRODUCT TRANSITION** : Factors required to transfer a product from one technology to another
- 

Maintainability
Flexibility
Testability

Portability
Reusability
Interoperability

**PRODUCT
REVISION**

**PRODUCT
TRANSITION**

PRODUCT OPERATION

Correctness ,Reliability,Usability,Integrity,Efficiency



Software Quality Assurance–SQA

Quality assurance is an essential activity for any business that produces products to be used by others.

- ▶ **SQA** – Planned and systematic pattern of actions that are required to ensure high quality in software.

- ▶ **SQA Activities are performed by 2 sets of people-----→**
 - A) **Software Engineers**– Who do technical work .
They address quality by -----?
 - 1) Applying solid technical methods and measures
 - 2) Conducting FTR
 - 3) Performing well planned s/w testing

 - B) **SQA group**–responsible for quality assurance planning ,oversight, record keeping analysis and reporting.

Software Quality Assurance–SQA

ACTIVITIES PERFORMED BY SQA GROUP

- 1) Prepare an SQA plan for a project.

SQA plan developed during project planning

Identifies -----→

1. Evaluations to be performed.
2. Audits and reviews to be performed.
3. Standards applicable to the project.
4. Procedures for error tracking.
5. Documents to be produced by the SQA group.
6. Amount of feedback provided to the s/w project team

Software Quality Assurance–SQA

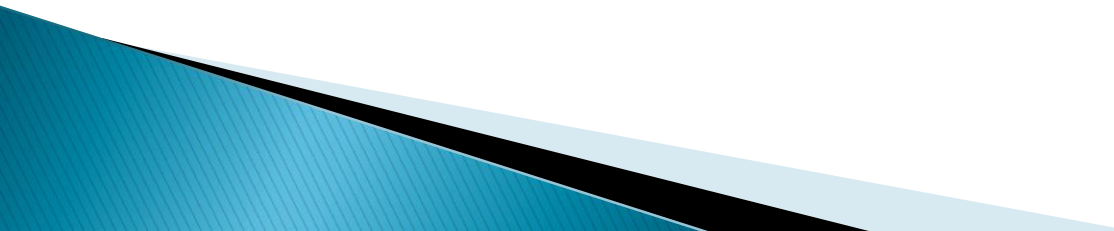
2. Participates in the development of the project's software process description.

The s/w team selects a process for the work to be performed.

The SQA group reviews process description for compliance with organization policy, internal software standards, externally imposed standards (ISO-9001)

3.Reviews s/w engineering activities to verify compliance with defined s/w process.

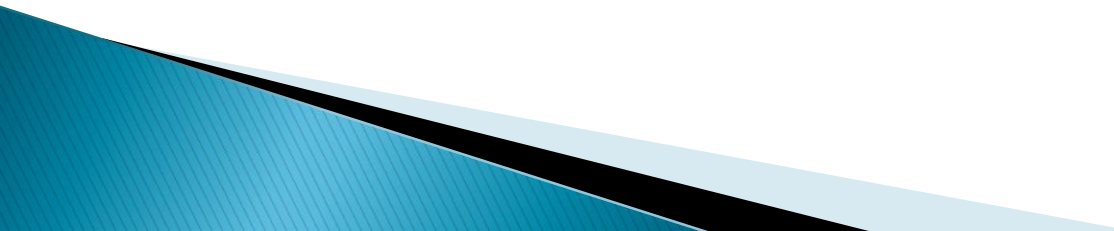
The SQA group identifies documents and tracks deviations from the process and verifies corrections that have been made.



Software Quality Assurance–SQA

4. Audits designated s/w work products to verify compliance with those defined as part of s/w process

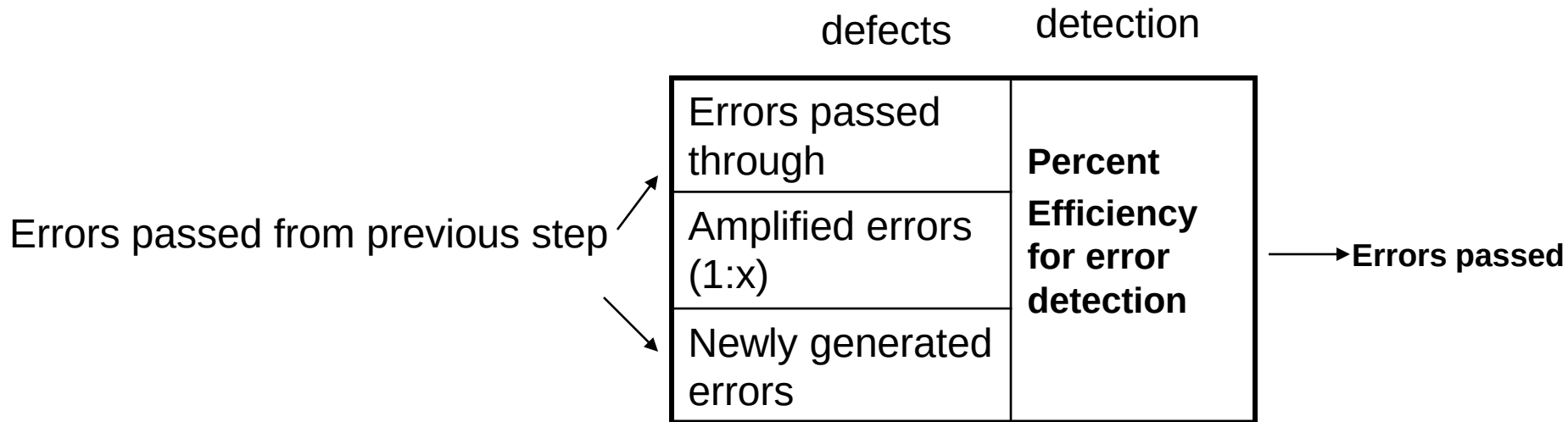
Includes periodic reporting of its work to the project manager.

5. Ensures that deviations in s/w work products are handled according to a documented procedure.
 6. Records any non compliance and reports to senior manager.
- 

Defect Amplification and Removal

- ▶ A defect amplification model is used to illustrate the generation and detection of errors during preliminary design, detail design and coding steps of the s/w engineering process.

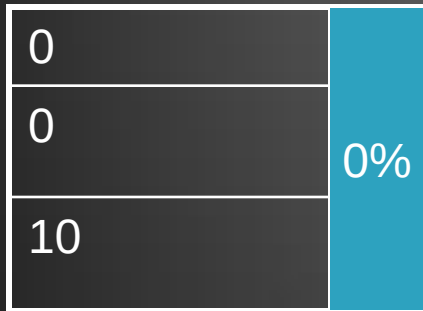
DEVELOPMENT STEP



Defect Amplification Model

Defect Amplification, no review

PRELIMINARY DESIGN

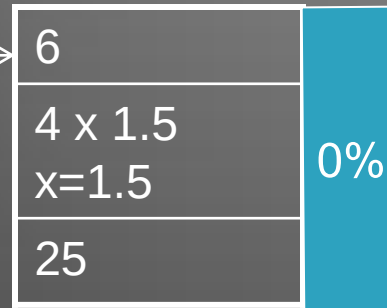


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DETAIL DESIGN

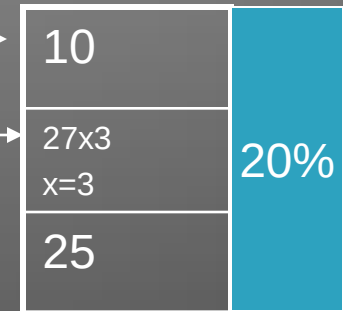


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27

Code/unit test

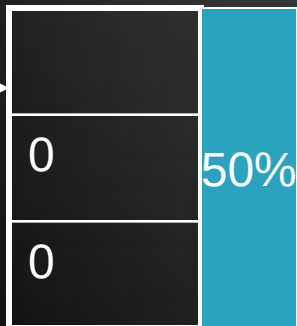


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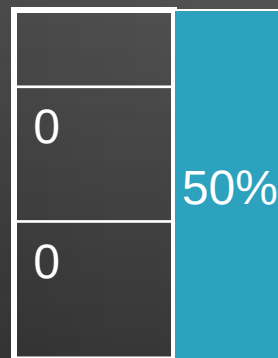
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Integration test



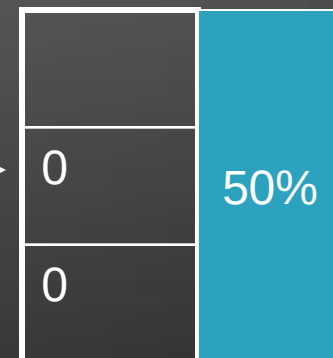
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Validation test



24

System test



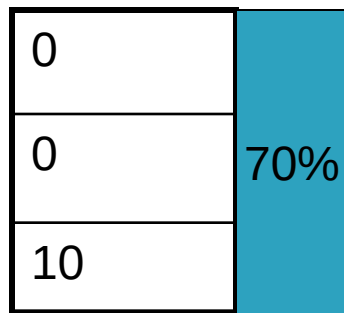
12



Design Amplification, reviews conducted

Contd.

Preliminary design

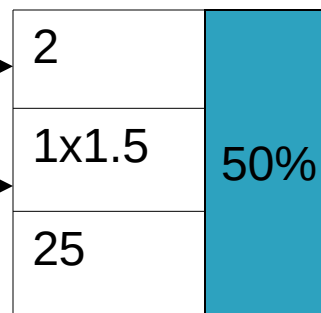


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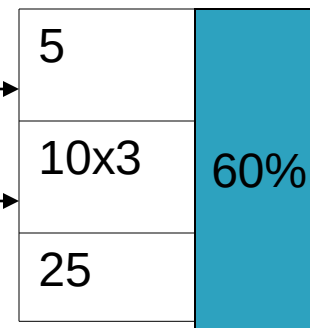
Detail design



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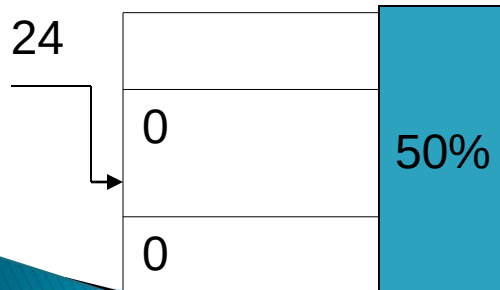
Code/unit test



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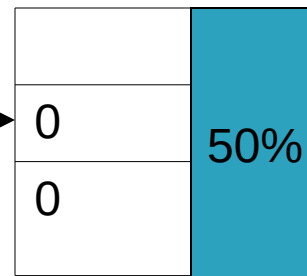
To integration

Integration test



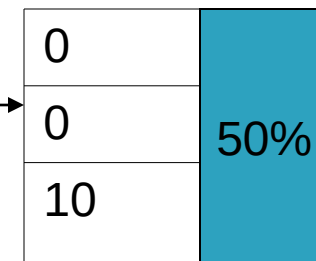
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Validation test



6

System test



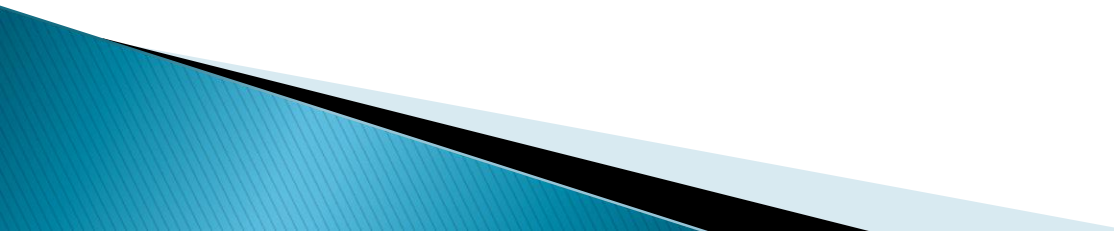
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Latent errors

Software Reviews

- ▶ Filters for the s/w engineering process
- ▶ Are applied at various points during s/w development and serve to uncover errors and defects that can be removed
- ▶ S/w reviews **purify** the s/w engineering activities i.e. analysis, design and coding

FORMAL TECHNICAL REVIEWS

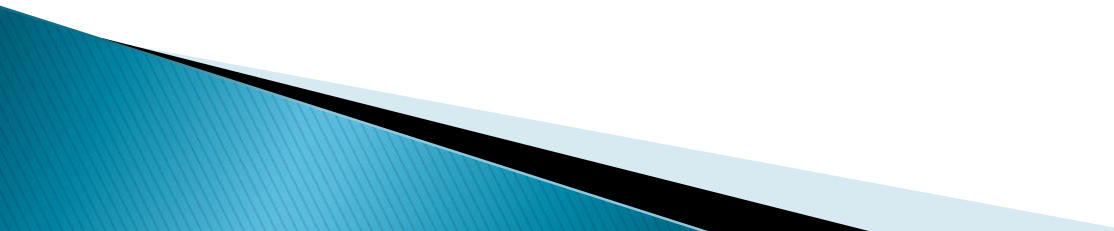
- ▶ SQA activity performed by s/w engineers
 - ▶ FTR is a class of reviews that includes walkthroughs inspections, round-robin reviews etc.
- 

Software Reviews

REVIEW MEETING

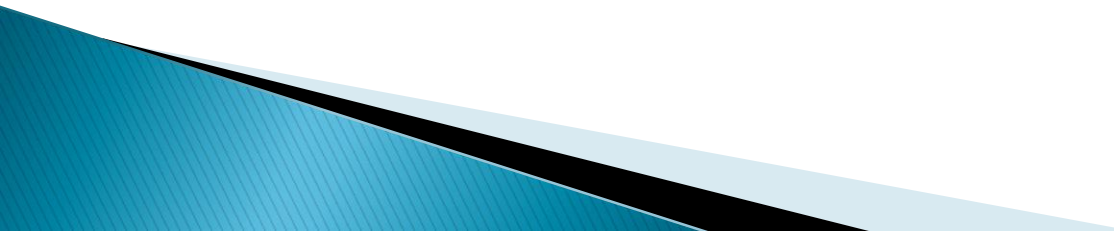
- ▶ Producer informs the project leader that the product is complete and that a review is required.
- ▶ Project leader contacts review leader.
- ▶ Review leader distributes the product materials to 2–3 reviewers for advance preparation,

Members of the team

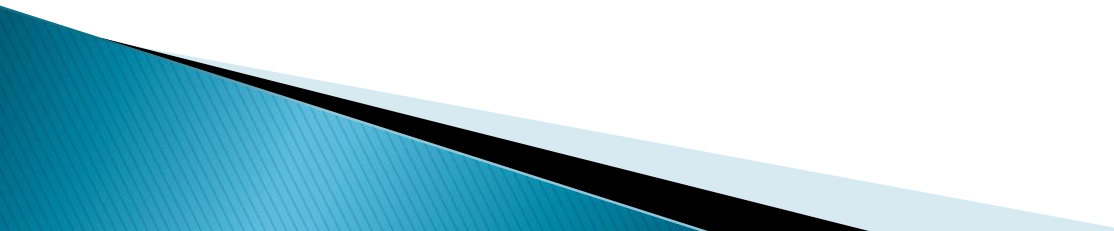
1. Review leader
 2. Review team
 3. Producer
- 

Software Reviews

Constraints of the meeting :

- 1) Only 2–3 people should be involved.
 - 2) Advance preparation requires not more than 2–3 hrs for each person.
 - 3) Duration of the meeting should be less than 2 hrs.
- 

Software Reviews

- ▶ During the meeting one of the reviewers ----→ recorder
 - ▶ FTR begins with a brief introduction of the agenda.
 - ▶ Producer proceeds to “walk through” the work product while reviewers raise issues based on their advance preparation.
 - ▶ Recorder notes each valid problem discovered.
 - ▶ At the end of the review all attendees decide ---→
 1. Accept product without modification.
 2. Reject the product due to severe errors(indicating another review is required when corrected)
 3. Accept the product with minor modification(no review required)
- 

Software Reviews

Review Reporting and Record keeping

Summarizes all issues raised during the review meeting.

A review summary report answers 3 questions---→

1. What was reviewed?
2. Who reviewed it?
3. What were the findings and conclusions?

RSR becomes part of the project's historical record.

Review guidelines :

1. Review the product and not the producer.
2. Set an agenda and maintain it.
3. Limit debate and rebuttal.
4. Emphasis should be placed on detecting errors and not correcting errors (during meeting)

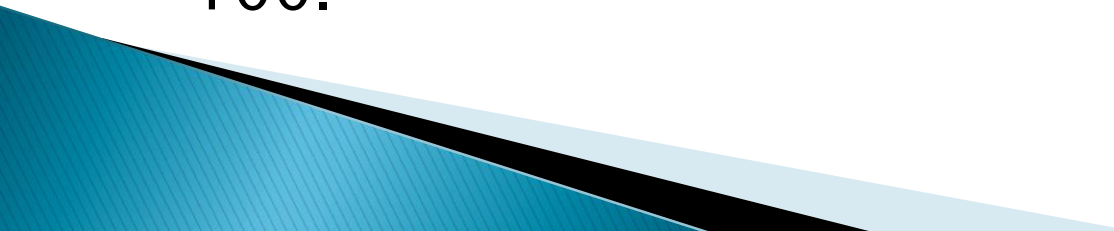
Software reliability

The probability of failure free operation of a computer program in a specified environment for a specified time.

Eg.

Program X is estimated to have a reliability of 0.96 over eight elapsed processing hours.

In other words, if program X were to be executed 100 times and require eight hours of elapsed processing time (execution time), it is likely to operate correctly (without failure) 96 times out of 100.



Software reliability

Measures for reliability:

MTBF(Mean Time Between Failures)

Parameters required to compute MTBF :

1. MTTF : (Mean Time To Fail)

Average time between two successive failures, observed over a large number of failures. To measure MTTF, we can record the failure data for n failures.

Let the failures occur at the time instants $t^1, t^2, \dots, t^n$. Then, MTTF can be calculated as

$$\sum_{i=1}^n \frac{t_{i+1} - t_i}{(n-1)} .$$

Software reliability

2. MTTR(Mean Time To Repair)

Once failure occurs, some time is required to fix the error.

MTTR measures the average time it takes to track the errors causing the failure and to fix them.

$$MTBF = MTTF + MTTR.$$

Software Availability

Software availability is the probability that a program is operating according to requirements at a given point in time.

Availability = uptime / (uptime + downtime)

Availability = $[MTTF / (MTTF + MTTR)] * 100$

Eg: A system going down for 1ms every hr. Find the availability.

$$\begin{aligned} &= (3600 - 0.001) / 3600 \\ &= 99.99 \% \end{aligned}$$